HIREENGINE: AI RECRUITMENT AGENT

A Mini Project Report submitted by

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to APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the degree of
Master of Computer Applications



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CERTIFICATE

This is to certify that the report titled **HireEngine: AI Recruitment Agent** is a bona-fide record of the work related to the paper 20MCA245 Mini Project done by

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Declaration

I, **Mahmood Ahsan K** , studying in Third Semester MCA (2025 admissions) class of Vidya Academy of Science & Technology, Thrissur – 680501, hereby declare that the project report ("HireEngine: AI Recruitment Agent") submitted by me is a bona fide work done by me under supervision of Mr. Manesh D. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place	: Thrissur – 680501	Signature of student	: .....
Date	: October 22, 2026	Name of student	: Mahmood Ahsan K

Acknowledgment

I wish to record my indebtedness and thankfulness to all who helped me prepare this Report titled " HireEngine: AI Recruitment Agent " and present it in a satisfactory way. This Report is part of my work related to the paper 20MCA245 Mini Project.

I am especially thankful to my guide and supervisor Mr. Manesh D in the Department of Computer Applications for giving me valuable suggestions and critical inputs in the preparation of this report. I am also thankful to Dr. Reji C Joy, the Head of Computer Applications Department for encouragement.

My friends in my class have always been helpful and I am grateful to them for patiently listening to my presentations on my work related to the Project.

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Synopsis

HireEngine is a web-based recruitment platform that organizes job postings, candidate applications, resume review, and interview rounds in a single workflow. Recruiters create a job, define required skills and interview availability, and share a public application link. Candidates submit their details and a PDF or DOCX resume through that link.

The first round processes the resume by extracting its text, removing contact details, applying AI-assisted anonymization, and producing a rubric-based score for recruiter review. Candidates selected by the recruiter can proceed to a timed AI voice interview. The interview transcript is evaluated against role-related criteria and shown as advisory information. Candidates who advance to the final round attend a scheduled video interview, where the recruiter records feedback and a selection decision. Automated email tasks and scheduled background work support the process, including cleanup of candidate application data after the configured 90-day retention period.

The application uses Django and Django REST Framework for its API, PostgreSQL for persistent data, Celery and Redis for background processing, and React for the browser interface. OpenAI services support resume processing and realtime voice interviews, while LiveKit provides video interview rooms. AI scores assist review; recruiters retain responsibility for advancement and final hiring decisions.

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PROJECT REPORT

Chapter 1

INTRODUCTION

1.1 Problem Definition

Recruitment information is often split among job boards, email, and spreadsheets. This fragmentation makes it harder to track applicants, compare qualifications consistently, coordinate interviews, and apply data-retention rules. Manual resume review can expose personal details before job-related qualifications are considered. HireEngine brings these tasks into one workflow and supports anonymized, rubric-based evaluations; recruiters retain responsibility for advancement and hiring decisions.

1.2 Objectives

The objectives are to provide secure recruiter accounts and job management; accept applications with consent and resumes; redact and score resume text; support CV review, AI voice, and HR video rounds; coordinate interviews; record feedback and outcomes; send notifications; and apply the candidate-data retention policy.

1.3 Purpose of the Project

The project gives recruiters one application for managing hiring from job creation to the final decision. Candidates apply and attend interviews through browser pages; recruiters review applications, control advancement, and record feedback. AI assists with resume and interview evaluation, while hiring decisions remain with the recruiter.

Chapter 2

SYSTEM ANALYSIS

2.1 The Existing System

In a conventional hiring process, job details, resumes, interview schedules, and feedback may be maintained in separate documents or services. Recruiters must manually update candidate status and coordinate available times, making the process difficult to audit and potentially inconsistent. Separate tools can also lead to repeated data entry, missed communications, and uncertainty about the status of a candidate. A centralized system can reduce this administrative fragmentation and make progress visible to the recruiter.

2.2 Proposed System

HireEngine provides a recruiter workspace and public candidate pages connected to a Django API. The system follows three rounds. In Round 1, a candidate submits an application and resume through a job-specific link. A background task extracts text from supported PDF or DOCX files, removes contact details, performs AI-assisted anonymization, and creates a rubric-based CV score. The recruiter reviews the applications and chooses which candidates to advance.

In Round 2, shortlisted candidates receive an invitation to a scheduled AI voice interview. A time-limited OpenAI Realtime session conducts a role-focused interview. The transcript is scored against technical, depth, communication, and relevance criteria and is made available to the recruiter as advisory information. In Round 3, selected candidates attend a LiveKit video interview with the recruiter. The recruiter records a rating, notes, and a selected or rejected decision. Email notifications, deadline checks, interview scheduling, and retention cleanup are supported by background tasks.

Chapter 3

FEASIBILITY STUDY

3.1 Technical Feasibility

HireEngine uses Python 3.12, Django 5.1, Django REST Framework, PostgreSQL 16, Celery 5, Redis 7, React 19, Vite, and Docker Compose. Its main services can be run with the provided container configuration. OpenAI, LiveKit, and SMTP credentials are needed for their corresponding external features. Camera and microphone access requires HTTPS or a browser localhost secure context.

3.2 Economical Feasibility

The backend, frontend, database, queue, and container technologies are available as open-source software. Production costs may include hosting, storage, email, OpenAI API usage, and LiveKit usage. Actual expenses depend on applicant and interview volumes and the selected service plans, so they should be estimated for the intended deployment.

3.3 Operational Feasibility

Recruiters use the dashboard to create jobs, review candidates, schedule interviews, and record decisions. Candidates apply and join interviews through public or private links. Users need a current browser and network access; interview rounds require microphone or camera permissions. AI evaluations are advisory, and recruiters retain control of advancement and hiring decisions.

Chapter 4

SYSTEM DESIGN

4.1 Input Design

The application accepts the following inputs through browser forms and API endpoints.

- **Recruiter registration:** full name, company name, work email, password, and optional company description, address, and timezone.
- **Job details:** job title, description, required skills, minimum experience, employment type, location, work mode, optional salary range, application deadline, and shortlist limits.
- **Candidate application:** full name, email, phone number, consent, and a resume in PDF or DOCX format.
- **Interview availability:** interview round and start and end times for the available scheduling window.
- **Recruiter evaluation:** advancement choice, interview feedback, rating, notes, and final decision.

4.2 Output Design

The application displays recruiter dashboard statistics, job stages, applications, CV-processing status, scores and score breakdowns, interview schedules, transcripts, feedback, and final decisions. Candidates receive application confirmation and round-invitation messages and can view the relevant application or interview information through public links. Error responses communicate invalid forms, duplicate applications, expired deadlines, or unavailable interview windows.

4.3 Database Design

HireEngine uses PostgreSQL. Recruiter accounts are represented by the custom User model. A recruiter may own multiple Job records. Each job may have applications, round records, and availability windows. An Application belongs to one job and stores candidate contact information, resume location, anonymized text, processing status, CV score, and pipeline status. An Interview is linked to a job, an application, and the responsible recruiter; an AIResult or Feedback record stores the relevant evaluation. EmailLog records notification delivery status.

The data model uses UUID primary keys for the principal records. Normalized email and phone numbers are unique per job, each job has a unique public form slug, and only one interview of each kind can be created per application. Database constraints validate shortlist sizes and timing values and prevent overlapping reservations for the same recruiter and interview kind. The PostgreSQL exclusion constraint uses the btree_gist extension.

Deleting a job cascades to its applications and related interview records. Candidate application data and associated resume files are removed by the scheduled cleanup task after the configured 90-day retention period for completed or removed jobs. Recruiter and candidate access is separated between authenticated recruiter endpoints and public application or token-based interview endpoints.

Chapter 5

IMPLEMENTATION, TESTING AND VALIDATION

5.1 Implementation

The backend is divided into Django applications for accounts, jobs, applications, interviews, notifications, and AI services. The React client communicates with the API through HTTP requests. Recruiter endpoints use JWT authentication; candidate application and interview endpoints are selectively public and use job slugs or interview tokens.

When a candidate submits a resume, the application is stored and asynchronous Celery tasks process the file and send a confirmation email. Text extraction supports PDF and DOCX documents and rejects unsupported or insufficiently text-based input. Contact details and URLs are removed before the anonymization and scoring steps. CV scoring uses skills, experience, and education categories with a total score out of 100. When OpenAI is unconfigured or unavailable, deterministic fallback routines are available for anonymization and scoring.

The AI interview service builds instructions from the job details and interview prompt and requests a short-lived OpenAI Realtime session credential. Once completed, the transcript is scored and stored with score details. The HR interview flow creates a LiveKit room token and allows the recruiter to record feedback. Celery workers perform resume and email tasks, while Celery Beat schedules recurring deadline and retention jobs. Environment variables are used for service configuration and credentials.

5.2 Testing Methods

The source repository includes Django tests for the main backend applications. The account tests cover email verification, OTP expiry, lockout after invalid attempts, resend cooldown, and login.

Job tests cover deadlines, ranking, shortlist capacity, invitations, and round transitions. Application tests cover public application submission, consent, supported resume formats, duplicate prevention, anonymization, and retention cleanup. Interview tests cover interview state and timing, AI results, video-room access, and recruiter feedback. Notification tests cover email task behavior.

The application tests use Django's test framework and the REST framework API client to exercise both successful and rejected requests. They check consent, resume formats and extracted text, duplicate contact details, and persisted application status, so the expected API behavior and stored data are both considered.

These test cases are present in the repository; this report does not claim a particular full-suite pass result.

5.3 Validation

Input validation is applied at the API serializer and model levels. Resume handling validates file type, readability, and minimum extracted text length. Database constraints protect unique application identifiers and interview scheduling rules. Authenticated recruiter views restrict job and application access to the owning recruiter, while public candidate actions use a job slug or interview token. Interview scores are constrained to their configured rubric ranges and are presented as advisory outputs. Recruiters make the final decisions.

Chapter 6

CONCLUSION

HireEngine centralizes job setup, application intake, resume processing, interview scheduling, recruiter feedback, and candidate outcome tracking. Its three-round workflow reduces fragmented administration by bringing application review, interview coordination, notifications, and decisions into one platform. PostgreSQL constraints help preserve data integrity, while Celery and Redis handle time-consuming background tasks and scheduled work.

The platform combines automated resume processing and AI-assisted interviews with recruiter review. The resulting scores are advisory: recruiters decide whom to advance and make the final hiring decision. The implemented data model, interview workflow, and retention task provide a foundation for a complete hiring process. Before production use, the external email, AI, video, storage, and database services require suitable configuration, monitoring, and operational controls.

The project demonstrates how these components can be combined while keeping recruiter decisions central. Its modular backend and browser-based interfaces provide a base for further evaluation with recruiters and candidates before wider deployment.

Chapter 7

SCOPE FOR FUTURE ENHANCEMENT

The project can be extended in several ways.

- **Audit trail:** record recruiter stage changes and decisions.
- **Score explanations:** show clearer evidence for CV and interview rubric scores.
- **Fairness evaluation:** assess anonymization and scoring with a consented, representative dataset.
- **Accessibility and localization:** add configurable rubrics, accessible candidate experiences, and additional languages.
- **Production operations:** improve deployment automation, secret management, monitoring, backups, and retention controls.
- **Expanded testing:** add end-to-end coverage for concurrent scheduling and external-service failures.

These improvements would make the hiring process more transparent and reliable as usage grows. They should be prioritized using recruiter and candidate feedback, while considering fairness, accessibility, and data protection requirements.

APPENDICES

Appendix A

SCRUM PROCESS ARTIFACTS

A.1 Product Backlog

The work items and effort estimates below are reconstructed from the implemented project features. The estimates are illustrative and are not recorded actual hours.

Priority	Product backlog items	User story	Estimate (Hours)
1	Recruiter accounts	As a recruiter, I want to manage hiring securely.	24
2	Job management	As a recruiter, I want to create and publish jobs for candidates.	32
3	Candidate applications	As a candidate, I want to submit my resume through a job link.	28
4	CV review	As a recruiter, I want anonymized resumes and scores for review.	40
5	AI voice interviews	As a candidate, I want to complete a scheduled voice interview.	40
6	HR video interviews	As a recruiter, I want to record interview feedback and final decisions.	40

A.2 Scrum Meeting Details

The dates below are illustrative sample dates based on the supplied project start month of July 2026. They are not verified meeting records. Dates use day/month format in 2026.

Sl. No.	Date	Sprint backlog	Product shippable Increment	Important Decisions
1	10/7	Account verification and API foundation	Recruiter registration and login	Use verified email and JWT authentication.
2	24/7	Jobs and candidate applications	Job form and public application flow	Validate consent and resume submissions.
3	14/8	Resume processing and CV review	Extracted, anonymized, scored CVs	Treat automated scores as advisory.
4	4/9	AI voice interview	Realtime interview and transcript scoring	Restrict interview to role-related questions.
5	25/9	HR video interviews and notifications	Scheduled LiveKit meetings and feedback	Prevent conflicting recruiter interview slots.
6	9/10	Testing and retention tasks	Test coverage and scheduled cleanup	Retain candidate data for the configured 90 days.

Appendix B

UML DIAGRAMS

B.1 Class Diagram

B.2 State Diagram

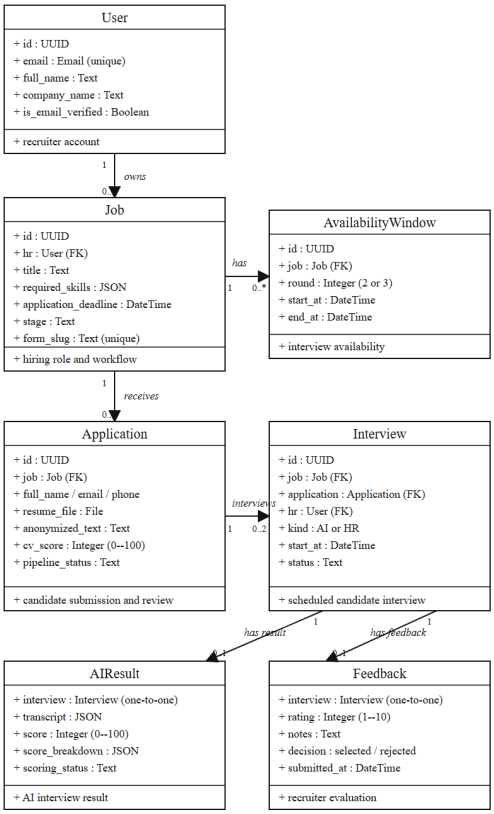


Figure B.1: Class Diagram of HireEngine

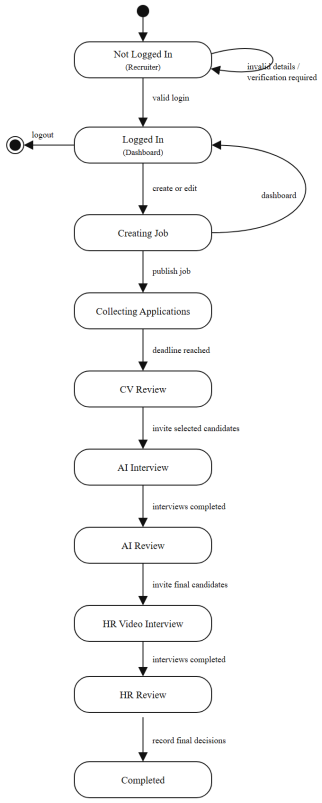


Figure B.2: State Diagram of HireEngine

Appendix C

TABLE STRUCTURE

Sl. No.	Field	Datatype	Constraint	Description
1	id	UUID	Primary Key	Recruiter ID
2	email	Email	Unique	Recruiter login
3	company_name	Text	Not Null	Recruiter company
4	is_email_verified	Boolean	Default false	Email verification status

Table C.1: User table

Sl. No.	Field	Datatype	Constraint	Description
1	id	UUID	Primary Key	Job ID
2	hr_id	UUID	Foreign Key	Job owner
3	title	Text	Not Null	Job title
4	required_skills	JSON	Not Null	Role skills
5	stage	Text	Not Null	Pipeline stage

Table C.2: Job table

Sl. No.	Field	Datatype	Constraint	Description
1	id	UUID	Primary Key	Application ID
2	job_id	UUID	Foreign Key	Applied job
3	resume_file	File	Not Null	Candidate resume
4	cv_score	Small Integer	0–100	CV evaluation
5	pipeline_status	Text	Not Null	Candidate stage

Table C.3: Application table

Sl. No.	Field	Datatype	Constraint	Description
1	id	UUID	Primary Key	Interview ID
2	application_id	UUID	Foreign Key	Related application
3	kind	Text	AI or HR	Interview type
4	start_at	DateTime	Not Null	Scheduled start
5	status	Text	Not Null	Interview state

Table C.4: Interview table

Appendix D

SAMPLE SCREENSHOTS

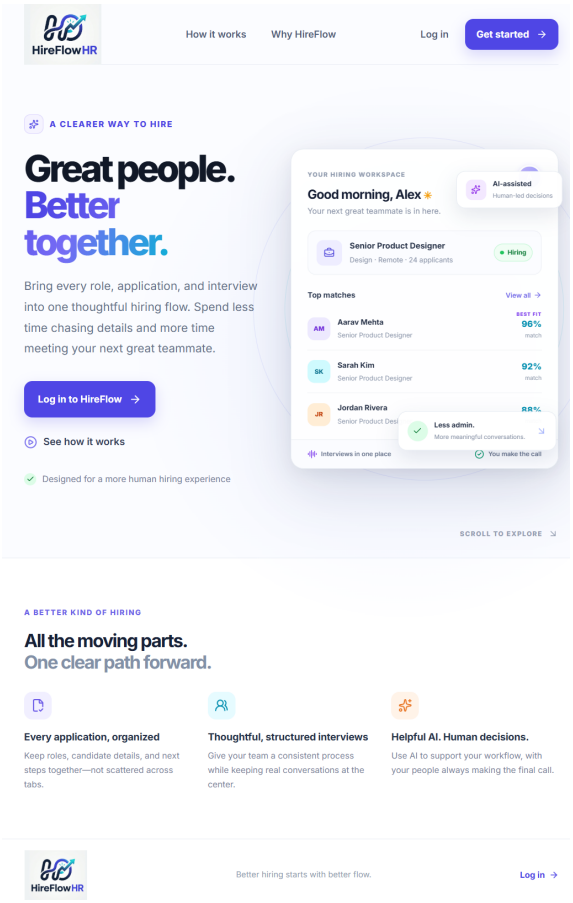


Figure D.1: Public landing page

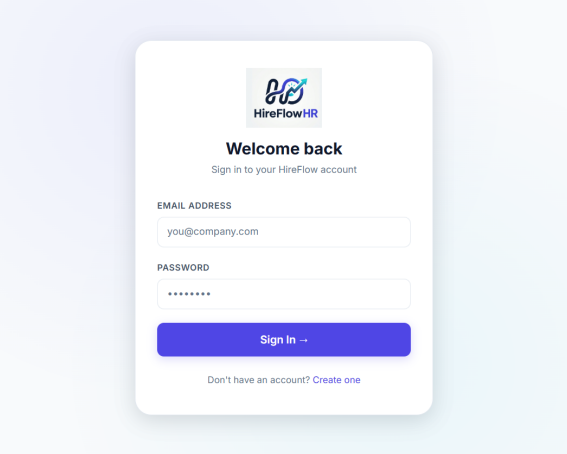


Figure D.2: Recruiter login page

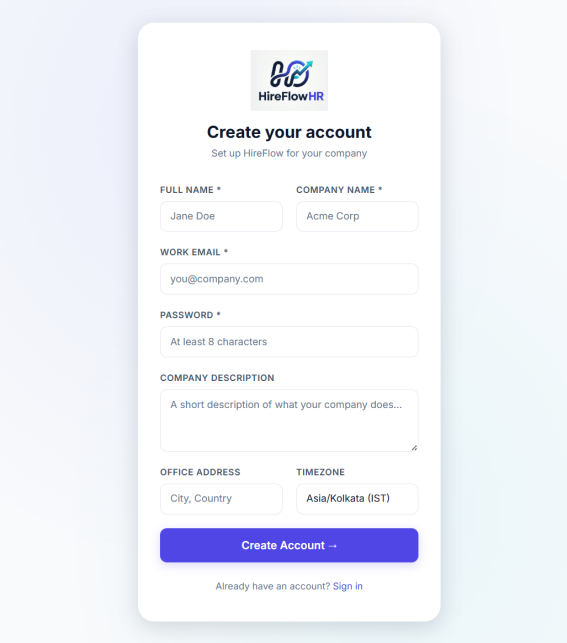


Figure D.3: Recruiter registration page

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